

Degrees of Uncomputability: Post's problem and the Finite Injury Priority method

Master's Degree in Computer Science

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Table of Contents

1 Introduction

► Introduction

► Degrees of uncomputability

► The Finite Extension method

► The Finite Injury Priority method



What is computation?

Defining computation

1931 – Kurt Gödel defines **General Recursive Functions** to prove his Incompleteness Theorems. It's the first mathematical definition of computation.

- *Constant functions*: return the natural number n
- *Successor function*: on input x return $x + 1$
- *Projection function*: on input $(x_1, \dots, x_n)$ return x_i
- *Composition operator*: on input $h, g_1, \dots, g_m$ return the function that applies h together on the outputs of the applications of $g_1, \dots, g_m$
- *Primitive recursion operator*: on input h, g , return the function that applies g if the first input is 0 and otherwise applies h on the first argument x_0 and then recursively applies the process on the predecessor of x .
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1931 – Kurt Gödel defines **General Recursive Functions** to prove his Incompleteness Theorems.

- *Constant functions:* $C_n^k(x_1, \dots, x_k) := n$ for all $k, n \in \mathbb{N}$
- *Successor function:* $S(x) := x + 1$
- *Projection function:* $P_i^k(x_1, \dots, x_k) := x_i$ for all $k \in \mathbb{N}$
- *Composition operator:*
 $(h \circ (g_1, \dots, g_m))(x_1, \dots, x_k) := h(g_1(x_1, \dots, x_k), \dots, g_m(x_1, \dots, x_k))$
- *Primitive recursion operator:* $\rho(g, h) := f$ where:

$$f(x_0, x_1, \dots, x_k) = \begin{cases} g(x_1, \dots, x_k) & \text{if } x_0 = 0 \\ h(y_0, f(y_0, x_1, \dots, x_k)) & \text{if } x_0 = S(y_0) \end{cases}$$

- *Minimization operator:* $\mu(f) := f'$ where:

$$f'(x_1, \dots, x_k) = \begin{cases} z & \text{if } f(z, x_1, \dots, x_k) > 0 \text{ and } f(i, x_1, \dots, x_k) > 0 \text{ for all } i < z \\ * & \text{otherwise} \end{cases}$$



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1935 – Alonzo Church defines **λ -calculus** to settle the Entscheidungsproblem problem.

- *Language grammar:* $M, N ::= x \mid \lambda x.M \mid MN$
- *Substitution:* $M[x := N]$ means that x gets substituted by N in M
- *α -rule:* $\lambda x.M \xrightarrow{\alpha} \lambda y.(M[x := y])$
- *β -rule:* $(\lambda x.M)N \xrightarrow{\beta} M[x := N]$
- *η -rule:* $\lambda x.Mx \xrightarrow{\eta} M$ if $x \notin \text{free}(M)$

Stephen Kleene proves that λ -calculus is equivalent to general recursive functions



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What is computation?

Defining computation

1936 – Alan Turing defines a simple abstract model of a computer.

- Infinite tape subdivided in cells with a read-write head
- Set of instructions defined by a function δ
- Each instruction moves the head left or right based on the current state and the value read.
- It may also change the value below the head before moving.

Turing's machine also settles the Entscheidungsproblem and it is equivalent to λ -calculus.
Recognized by Gödel himself as the best definition of computation.



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$\Phi_{i,s}(x)$ denotes the **computation** of the i -th TM for $s \in \mathbb{N}$ steps on input $x \in \mathbb{N}$.

- $\Phi_{i,s}(x) \downarrow$: the computation halts after s steps
- $\Phi_i(x) \downarrow$: there is a $s \in \mathbb{N}$ such that $\Phi_{i,s}(x) \downarrow$
- $\Phi_i(x) \uparrow$: there is no $s \in \mathbb{N}$ such that $\Phi_{i,s}(x) \downarrow$
- $\psi_i(x)$: output of the computation (defined iff $\Phi_i(x) \downarrow$)



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Computability

Definitions

Given a set $S \subseteq \mathbb{N}$, we say that S is:

- **Semi-decidable** when $\exists i \in \mathbb{N}$ such that $\forall x \in \mathbb{N}$ it holds that $\psi_i(x) = 1$ if $x \in S$, otherwise $\psi_i(x) = 0$ or $\Phi_i(x) \uparrow$.
- **Decidable** when $\exists i \in \mathbb{N}$ such that $\forall x \in \mathbb{N}$ it holds that $\psi_i(x) = 1$ if $x \in S$, otherwise $\psi_i(x) = 0$.
- **Recursively enumerable** when there is an algorithmic procedure $\mathcal{A} : \mathbb{N} \rightarrow \{0, 1\}$ such that $S = \{A(0), A(1), A(2), \dots\}$

Obs. 1: S is semi-decidable if and only if it is recursively enumerable

Obs. 2: S is decidable if and only if both S and $\bar{S}$ are semi-decidable.



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Introduction

Turing's work

Turing [Tur36] proved that:

- Some sets that are semi-decidable but undecidable (e.g. $H = \{\langle i, x \rangle \mid \Phi_i(x) \downarrow\}$).
- Some sets cannot be semi-decided (e.g. $\overline{H}$).

Three degrees of computability: **solvable** problems, **semi-solvable** problems and **unsolvable** problems.

Are there some other degrees of computability?



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Table of Contents

2 Degrees of uncomputability

- ▶ Introduction
- ▶ Degrees of uncomputability
- ▶ The Finite Extension method
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Degrees of uncomputability

Turing degrees

Post [Pos44] formalized the idea of computability degrees through **Turing reductions**.

Let Φ_i^B denote the Turing machine Φ_i with access to an oracle machine that decides B .

- *Turing reducibility*: $A \leq_T B$ when $\exists i \in \mathbb{N}$ such that $\Phi_i^B(x) \downarrow$ for all $x \in \mathbb{N}$ and $A = \{x \in \mathbb{N} \mid \psi_i^B(x) = 1\}$.
- *Turing equivalence*: $A \equiv_T B$ when $A \leq_T B$ and $B \leq_T A$
- The set $\mathcal{D} = 2^{\mathbb{N}} / \equiv_T$ is the set of **Turing degrees**.
- $[A]$ is the class containing A over $\mathcal{D}$.



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Degrees of uncomputability

Turing degrees

We say that $[A]$ is **lower** than $[B]$, written as $[A] \preceq [B]$, if $A \leq_T B$.

$\preceq$ forms an ordering over the set $\mathcal{D}$, giving an hierarchy of unsolvability degrees.

$0 = [\emptyset]$ is the unique degree containing all decidable problems

- There is no degree below 0
- 0 is lower than any degree



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Degrees of uncomputability

Turing degrees

We say that $[A]$ is **lower** than $[B]$, written as $[A] \preceq [B]$, if $A \leq_T B$.

$\preceq$ forms an ordering over the set $\mathcal{D}$, giving an hierarchy of unsolvability degrees.

$0 = [\emptyset]$ is the unique degree containing all decidable problems

- There is no degree below 0
- 0 is lower than any degree



Degrees of uncomputability

The jump operator

The strict relation $[A] \prec [B]$ between degrees can be easily forced through **Turing jumps**.

- Given a set $X \subseteq \mathbb{N}$, the *Turing jump* of X is the set $X' = \{\langle i, x \rangle \mid \Phi_i^X(x) \downarrow\}$
 - $X \leq_T X'$ but $X' \not\leq_T X$ since X' is a variant of the Halting problem using X as an oracle.
- The jump of $\emptyset$ is exactly the Halting problem H
- We define $0'$ as the class of H (or $\emptyset'$)



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Degrees of uncomputability

Low degrees

- The jump operator can be iteratively applied to get infinite levels of the hierarchy
- A degree is said to be r.e. if it contains an r.e. set
- Not all sets inside an r.e. degree are r.e. (e.g. $\overline{H} \in 0'$)
- All r.e. degrees are below $0'$
 - Equivalently, all degrees above $0'$ are not r.e.
- Degrees below $0'$ are referred to as **low degrees**



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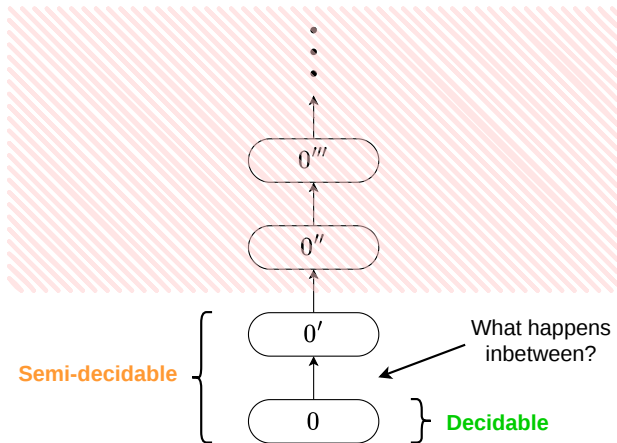
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Degrees of uncomputability

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Degrees of uncomputability

Post's problem

- In [Pos44], Emil Post investigates the existence of a degree d such that $0 \prec d \prec 0'$
- He was able to prove that for each degree $A \neq 0$ there is another degree B that is incomparable with A .
 - He was also able to force that A and B are low set
 - This answers his original question
- However, he couldn't find a way to force A and B to be r.e.

Post's problem: is there a r.e. degree d such that $0 \prec d \prec 0'$?



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Table of Contents

3 The Finite Extension method

- ▶ Introduction
- ▶ Degrees of uncomputability
- ▶ **The Finite Extension method**
- ▶ The Finite Injury Priority method



The Finite Extension method

Towards the solution to Post's problem

- Post's problem was solved 12 years later independently by Friedberg [Fri57] and Muchnik [Muc56] through the **finite injury priority method**.
- The method is an improvement on the **finite extension method** developed by Post in his original works
- Since the FIP method involves constructions that are way more complex than those of the FE method, we'll first give an example of the latter



The Finite Extension method

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The Finite Extension method

The idea

- We define a list of requirements $\{R_i\}_{i \in \mathbb{N}}$ that will enforce our desired property over the sets A and B
- We construct two list of strings $\{\alpha_i\}_{i \in \mathbb{N}}$ and $\{\beta_i\}_{i \in \mathbb{N}}$ where:
 - α_i is a finite extension of α_{i-1} (same goes for β_i and β_{i-1})
 - The strings α_i, β_i will satisfy requirement R_i
 - Strings are to be considered as partial functions from $\mathbb{N}$ to $\{0, 1\}$ (equivalently, as an infinite tape over $\{0, 1, *\}$)
- The sets $A = \bigcup_{i=0}^{+\infty} \mathbb{N}[\alpha_i]$ and $B = \bigcup_{i=0}^{+\infty} \mathbb{N}[\beta_i]$ will have our desired property
 - $\mathbb{N}[s] = \{x \in \mathbb{N} \mid s(x) = 1\}$ is the set of all natural numbers "selected" by s



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The Finite Extension method

The Kleene-Post theorem

[KP54] **Theorem:** There are two low sets A and B that are incomparable

- Requirements:

$$R_{2j} := (\psi_j^A \neq B)$$

$$R_{2j+1} := (\psi_j^B \neq A)$$

- Start by setting $\alpha_0 = \beta_0 = \varepsilon$



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- Consider any index $2j > 0$. Pick any position $x \in \mathbb{N}$ such that $\beta_{2j-1}(x) = *$ (guaranteed to exist)
- If there is any finite extension α' of α_{2j-1} such that $\Phi_{2j-1}^{\{\alpha'\}}(x) \downarrow$, we set:

$$\alpha_{2j} = \alpha' \quad \beta_{2j}(y) = \begin{cases} 1 - \psi_{2j}^{\{\alpha'\}}(x) & \text{if } y = x \\ \beta_{2j-1}(y) & \text{otherwise} \end{cases}$$

- Otherwise, we set:

$$\alpha_{2j} = \alpha_{2j-1} \quad \beta_{2j}(y) = \begin{cases} \text{rand}(\{0, 1\}) & \text{if } y = x \\ \beta_{2j-1}(y) & \text{otherwise} \end{cases}$$



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The Finite Extension method

The Kleene-Post theorem

Is there a way to set a finite number of these bits
in a way such that $\Phi_{2j}^{\alpha'}(x)$ terminates?

α_{2j-1}	0	1	0	1	*	*	*	1	*	1	*	*	*	*	*	*	...
-----------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

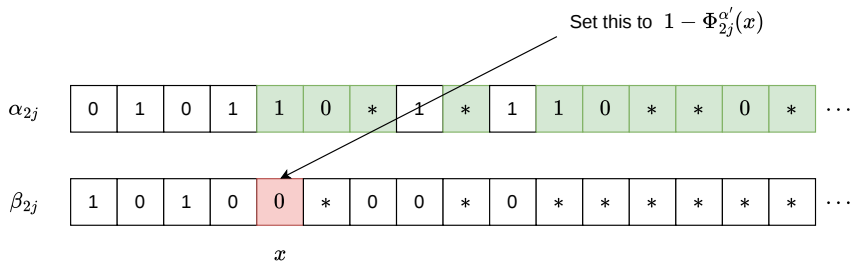
β_{2j-1}	1	0	1	0	*	*	0	0	*	0	*	*	*	*	*	*	...
----------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

x



The Finite Extension method

The Kleene-Post theorem





The Finite Extension method

The Kleene-Post theorem

- Strings α_{2j}, β_{2j} satisfy requirement R_{2j}
- Symmetrical idea applied to index $2j + 1 > 0$, satisfying R_{2j+1}
- **Incomparability:** all requirements are satisfied
- **Lowness:** A and B can be enumerated through an oracle for H but not without it





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The Finite Extension method

Strenghts and weaknesses

- We observe that this general method can be easily modified and extended.
- For instance,

$$R_{2j} := (\text{If } W_j \text{ infinite then } W_j \cap A \neq \emptyset) \qquad R_{2j+1} := (\psi_j^B \neq A)$$

imposes that A is a **simple set** and that B is a low set such that $A \not\leq_T B$

- $W_j = \{x \in \mathbb{N} \mid \Phi_j(x) \downarrow\}$
- A is simple when A is r.e., $\bar{A}$ is infinite and $\bar{A}$ doesn't contain an infinite r.e. set



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Table of Contents

4 The Finite Injury Priority method

- ▶ Introduction
- ▶ Degrees of uncomputability
- ▶ The Finite Extension method
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The Finite Injury Priority method

The idea

- **Injury:** a requirement gets injured when its satisfaction is not guaranteed anymore.
- The key idea is to assign a priority level to each requirement and force two conditions during the construction of the sets:
 1. Each requirement has finitely many requirements with higher priority
 2. Each requirement can be injured only by requirements with higher priority
- We'll use sets instead of strings to construct A and B



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The Finite Injury Priority method

The query function

- To keep track of injuries, we'll use a **query function** $\omega_{i,s}^A(x)$ that returns the largest index queried by $\Phi_{i,s}^A(x)$.

$$\omega_{i,s}^A(x) = \begin{cases} \max\{z \in \mathbb{N} \mid z \text{ gets queried in } \Phi_{i,s}^A(x)\} & \text{if } \Phi_{i,s}^A(x) \downarrow \\ -1 & \text{otherwise} \end{cases}$$

where $A \subseteq \mathbb{N}$ and $i, s, x \in \mathbb{N}$.



The Finite Injury Priority method

The Friedberg-Muchnik theorem

[Fri57; Muc56] **Thm.:** There are two r.e. sets A and B that are incomparable

- Requirements:

$$R_{2j} := (\psi_j^A \neq B)$$

$$R_{2j+1} := (\psi_j^B \neq A)$$

- Start by setting $A_0 = B_0 = \emptyset$
- We say that an index x is a **witness** for the requirement R_{2i} at step s if $\Phi_{i,s}^{B_s}(x) \downarrow$ and $\psi_{i,s}^{B_s}(x) \neq A_s(x)$ (symmetric definition for R_{2i+1})



The Finite Injury Priority method

The Friedberg-Muchnik theorem

[Fri57; Muc56] **Thm.:** There are two r.e. sets A and B that are incomparable

- Requirements:

$$R_{2j} := (\psi_j^A \neq B)$$

$$R_{2j+1} := (\psi_j^B \neq A)$$

- Start by setting $A_0 = B_0 = \emptyset$
- We say that an index x is a **witness** for the requirement R_{2i} at step s if $\Phi_{i,s}^{B_s}(x) \downarrow$ and $\psi_{i,s}^{B_s}(x) \neq A_s(x)$ (symmetric definition for R_{2i+1})



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- On each step $s \in \mathbb{N}$, for each $j \in \mathbb{N}$ we compute:
 - A witness $w_{j,s}$ for R_j at step s
 - A restriction index $r_{j,s}$ to fix the priority of the requirements
- **Invariants:**
 - $w_{j,s} = r_{j,s} = -1$ when no witness is known for R_j at step s
 - Indices that come before $r_{j,s}$ don't break any requirement.
- R_j gets *injured* at step s if $A_{s+1} - A_s$ contains some $x \leq r_{j,s}$.



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- Let $A_0 = B_0 = \emptyset$ and let $w_{j,0} = r_{j,0} = -1$ for each $j \in \mathbb{N}$.
- Consider a generic step $s = 2i$ (symmetrical argument for $s = 2i + 1$).
- For each $j \leq s$, if $w_{j,s-1} \neq -1$ we propagate the previous values because R_{2i} already has a witness.
 - $A_s = A_{s-1}$ and $B_s = B_{s-1}$
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- Suppose now that $w_{2h,s-1} = -1$.
- Let x be the minimum index such that $x \notin A_{s-1}$ and such that for all $k < 2h$ it holds that $x > r_{k,s}$.
- If $\Phi_{h,s-1}^{B_{s-1}}(x) \uparrow$, we preserve that $w_{2h,s} = r_{2h,s} = -1$ and propagate $A_s = A_{s-1}$, $B_s = B_{s-1}$. This trivially satisfies the requirement R_{2h} .
- Otherwise, if $\psi_{h,s-1}^{B_{s-1}}(x) = 0$, we set:

$$w_{2h,s} = x \qquad r_{2h,s} = \max(x, \omega_{h,s-1}^{B_{s-1}}(x))$$

$$A_s = A_{s-1} \cup \{x\} \qquad B_s = B_{s-1}$$



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- Suppose now that $w_{2h+1,s-1} = -1$.
- Let x be the minimum index such that $x \notin B_{s-1}$ and such that for all $k < 2h + 1$ it holds that $x > r_{k,s}$.
- If $\Phi_{h,s-1}^{A_{s-1}}(x) \uparrow$, we preserve that $w_{2h+1,s} = r_{2h+1,s} = -1$ and propagate $A_s = A_{s-1}$, $B_s = B_{s-1}$. This trivially satisfies the requirement R_{2h+1} .
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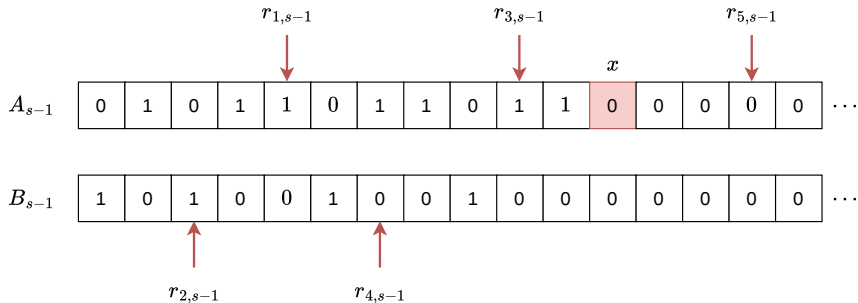
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What happens when $w_{4,s} = -1$?

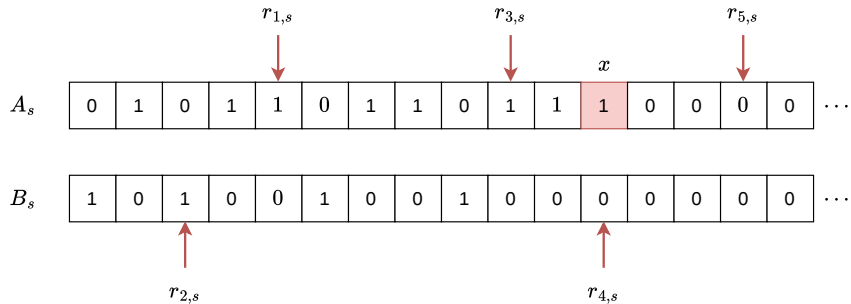




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Claim: For each $j \in \mathbb{N}$ it holds that:

1. R_j is injured a finite number of times
2. There is a step s_0 such that for all $s \geq s_0$ and for all $k < j$ no new index is added to A_s

Proof.

- The requirement R_j is injured at step s if some $x \leq r_{j,s}$ is added to A_{s-1} (or to B_{s-1}), which happens only when $w_{j,k} = -1$ and $\psi_{j,s}^{B_s}(x) = 0$ (or $\psi_{j,s}^{A_s}(x) = 0$)
- In other words, statement 2. of the claim follows from statement 1. by construction.



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The Friedberg-Muchnik theorem

- Let $g_{j,s} = \sum_{\substack{k < j \text{ s.t.} \\ w_{k,s} \neq -1}} 2^{-(k+1)}$ be the injure value of R_j at step s
- When R_j is injured at step $s - 1$, the smallest value $k < j$ such that $w_{k,s-1} \neq w_{k,s}$ must satisfy $w_{k,s-1} = -1$ and $w_{k,s} \neq -1$ by construction. Thus:

$$g_{j,s} - g_{j,s-1} \geq 2^{-(j+1)} - \sum_{k < h < j} 2^{-(h+1)} = 2^{-j}$$

- Hence, for each step s we have that $0 \leq g_{j,s} \leq 1 - 2^{-j}$, concluding that R_j can be injured at most $2^j - 1$ times



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- We observe that there is always a large enough step s_0^* satisfying statements 1. and 2. of the Claim.
- This step guarantees that for each $j \in \mathbb{N}$ it holds that $\lim_{s \rightarrow +\infty} w_{j,s}$ and $\lim_{s \rightarrow +\infty} r_{j,s}$ exist and are finite.
 - If for some $s' \geq s_0^*$ it holds that $w_{j,s'} \neq -1$ then it will hold from s' onward
 - Otherwise, there is a minimum value x_s^* such that $x_s^* \notin A_s$ and such that $x_s^* > r_{k,s}$ for all $k < j$
 - By construction $\Phi_{j,s}^{B_s}(x) \uparrow$ holds in this case



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- $A = \bigcup_{s=0}^{+\infty} A_s$ and $B = \bigcup_{s=0}^{+\infty} B_s$
- The claim implies that every requirement will be, eventually, satisfied by the construction.
- Moreover, the use of the restriction values $r_{j,s}$ allows us to recursively enumerate the sets A and B by restricting our interest to the indexes between 0 and $r_{j,s}$ for each $j \in \mathbb{N}$
 - This implies that A and B are both r.e.!





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Other interesting results

- **Infinite Injury Priority Argument:** Sacks [Sac64] proved that the above construction can be extended to a countably infinite argument
- **Density of r.e. sets:** For each pair of r.e. sets A, B there is another r.e. set C such that $A <_T C <_T B$
- Simpson [Sim77] proved that the first-order theory of $\mathcal{D}$ over the language $(\preceq, =)$ is many-one equivalent to the theory of true second-order arithmetic.



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Thank you for listening!
Any questions?



Bibliography

- [Fri57] Richard M. Friedberg. “Two recursively enumerable sets of incomparable degrees of unsolvability (solution of Post’s problem, 1944)”. In: (1957).
- [KP54] Stephen C. Kleene and Emil L. Post. “The upper semi-lattice of degrees of recursive unsolvability”. In: (1954).
- [Muc56] Albert A. Muchnik. “On the unsolvability of the problem of reducibility in the theory of algorithms”. In: (1956).
- [Pos44] Emil L. Post. “Recursively enumerable sets of positive integers and their decision problems”. In: (1944).
- [Sac64] Gerald E. Sacks. “The Recursively Enumerable Degrees are Dense”. In: (1964).
- [Sim77] Stephen G. Simpson. “First-Order Theory of the Degrees of Recursive Unsolvability”. In: (1977).
- [Tur36] Alan M. Turing. “On Computable Numbers, with an Application to the Entscheidungsproblem”. In: (1936).